

Shuo SUN

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Education

Doctor of Philosophy Candidate | National University of Singapore | 2021 - 2025

- Research Interests: Robotics, Autonomous Driving, Motion Prediction and Planning, Simulation
- Supervisor: Professor Marcelo H. Ang Jr. (Advanced Robotics Centre)

Bachelor of Engineering (Honours) | National University of Singapore | 2016 - 2020

- Double Major in Mechanical Engineering (ME) and innovation & Design Programme (iDP)
- Science & Technology Undergraduate Scholarship (2016 – 2020)

Work Experience

Graduate Tutor & Project Engineer | NUS Advanced Robotics Centre | Aug 2020 – Present

- As the team lead, leading the technical design & development of the **Autonomous Road Sweeper** project and the **Autonomous Mini-bus** project (details in the **Projects & Competitions** section).
- Designed & developed a full-stack autonomous driving software for semi-structured environment.
- Supervised 30+ Master's and 40+ Bachelor Thesis Projects.

Industrial Projects

Autonomous Mini-bus | NUS, Tron-E, Turing Drive Taiwan | Apr 2023 – Present

- As the project lead, developing an Autonomous Minibus for shuttle services on-demand.
- Designing the sensor configuration and hardware layout for a Tron-E new electric minibus model.
- Developing a complete autonomous driving software stack (including perception, HD map, localization, planning, control) for shuttle services in geofenced areas.

Autonomous Road Sweeper | NUS, Kärcher Germany | Apr 2020 – Apr 2023

- As the project lead, developed an Autonomous Road Sweeper prototype for professional cleaning.
- Retrofitted a Kärcher KM 105/110 R Bp Vacuum sweeper to a Drive-by-Wire vehicle.
- Designed & developed a complete autonomous driving software stack (including perception, HD map, localization, planning, control) for cleaning operations in a mixed indoor and outdoor environment.
- Demonstrated the prototype vehicle to Kärcher CTO and external customers.
- Transferred the prototype and software system to Kärcher Robotics team for commercialization.

Autonomous Golf Buggies | Micron Singapore | Jan 2019 – Dec 2019

- Led a 7-intern team, designed and developed the entire autonomous driving system for a fleet of golf buggies operating in Micron Singapore Fab.
- Designed and developed the Drive-by-Wire (DBW) system and autonomous driving hardware system on golf buggies from scratch.

- Implemented the object detection pipeline in the perception stack.
- Designed a map representation format to represent road networks in the HD map module.
- Implemented and improved a trajectory planning and Model Predictive Control (MPC) controller.
- Demonstrated our vehicles to the Deputy Prime Minister of Singapore and the Micron CEO.

Skills & Abilities

- **Robotics:** Autonomous Driving, Object Detection, Sensor Fusion, Motion Prediction, Motion Planning, Motion Control, Scenario Generation & Simulation.
- **Deep Learning:** CNNs, Transformers, Diffusion Models, LLMs.
- **Programming:** C/C++, Python, PyTorch, TensorFlow, ROS, OpenCV, PCL, Qt, MATLAB.

Highlighted Publications

- [1] **S. Sun**, Z. Gu, T. Sun, J. Sun, C. Yuan, Y. Han, D. Li, and M. H. Ang, “Drivescenegen: Generating diverse and realistic driving scenarios from scratch,” *IEEE Robotics and Automation Letters (RA-L 2024 & ICRA 2025)*, vol. 9, no. 8, pp. 7007-7014, Aug 2024.
- [2] **S. Sun**, J. Chen, J. Sun, C. Yuan, Y. Li, T. Zhang, and M. H. Ang, “Fiss+: Efficient and focused trajectory generation and refinement using fast iterative search and sampling strategy,” in the 2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS 2023**), Oct 2023.
- [3] **S. Sun**, Z. Liu, H. Yin, and M. H. Ang, “Fiss: A trajectory planning framework using fast iterative search and sampling strategy for autonomous driving,” *IEEE Robotics and Automation Letters (RA-L 2022)*, vol. 7, no. 4, pp. 9985–9992, Oct 2022.
- [4] **S. Sun**, T. Zhang, Z. Xiang, Y. Han, D. Li, J. Li, Z. Liu, and M. H. Ang, “Autonomous research platform for cleaning operations in mixed indoor & outdoor environments,” in The 17th International Conference on Control, Automation, Robotics and Vision (**ICARCV 2022**), Dec 2022.
- [5] H. Yang, S. Ding, J. Wang, **S. Sun**, R. Swaminathan, S. W. L. Ng, X. Pan, and G. W. Ho, “Computational design of ultra-robust strain sensors for soft robot perception and autonomy,” **Nature Communications**, vol. 15, no. 1, p. 1636, 2024.
- [6] J. Sun, J. Li, T. Liu, C. Yuan, **S. Sun**, Z. Huang, A. Wong, K. P. Tee, and M. H. Ang, “RMP-YOLO: A Robust Motion Predictor for Partially Observable Scenarios even if You Only Look Once,” in the 2025 IEEE International Conference on Robotics and Automation (**ICRA 2025**), arXiv preprint *arXiv:2409.11696*.
- [7] J. Sun, J. Li, T. Liu, C. Yuan, **S. Sun**, Y. Han, A. Wong, K. P. Tee, and M. H. Ang, “RMP: 3rd place solution for 2024 waymo open dataset challenge - motion prediction,” in Proceedings of the **CVPR 2024 Workshop** on Autonomous Driving, Seattle, WA, USA, Jun. 2024.
- [8] C. Yuan, Z. Zhang, J. Sun, **S. Sun**, Z. Huang, C. D. W. Lee, D. Li, Y. Han, A. Wong, K. P. Tee, and M. H. Ang, “DRAMA: An Efficient End-to-end Motion Planner for Autonomous Driving with Mamba,” arXiv preprint *arXiv:2408.03601*.
- [9] J. Li, T. Shen, Z. Gu, J. Sun, C. Yuan, Y. Han, **S. Sun**, and M. H. Ang, “Adm: Accelerated diffusion model via estimated priors for robust motion prediction under uncertainties,” arXiv preprint *arXiv:2405.00797*.
- [10] J. Sun, C. Yuan, **S. Sun**, S. Wang, Y. Han, S. Ma, Z. Huang, A. Wong, K. P. Tee, and M. H. Ang, “Controlmtr: Control-guided motion transformer with scene-compliant intention points for feasible motion prediction,” arXiv preprint *arXiv:2404.10295*.

- [11] Y. Han, Z. Liu, **S. Sun**, D. Li, J. Sun, Z. Hong, and M. H. Ang, "Carla-loc: Synthetic slam dataset with full-stack sensor setup in challenging weather and dynamic environments," arXiv preprint *arXiv:2309.08909*.
- [12] T. Zhang, **S. Sun**, J. Shi, S. Chen, M. H. Ang, J. Xin, and N. Zheng, "Maximum entropy inverse reinforcement learning based on frenet frame sampling for human-like autonomous driving," in 2023 IEEE 26th International Conference on Intelligent Transportation Systems (ITSC 2023), Sep 2023.
- [13] J. Sun, T. Liu, C. Yuan, **S. Sun**, A. Yong, K. P. Tee, and M. H. Ang, "An adaptive local context extraction method for motion prediction and planning," in the 10th IEEE International Conference on Cybernetics and Intelligent Systems, and Robotics, Automation and Mechatronics (CIS-RAM 2023).
- [14] J. Sun, C. Yuan, **S. Sun**, Z. Liu, T. Goh, A. Wong, K. P. Tee, and M. H. Ang, "Get-dipp: Graph-embedded transformer for differentiable integrated prediction and planning," In 2023 3rd International Conference on Computer, Control and Robotics (ICCCR), pp. 414-421, IEEE, 2023.
- [15] Y. Chen, **S. Sun**, H. Yin, and M. H. Ang, "Exploring the effect of 3d object removal using deep learning for lidar-based mapping and long-term vehicular localization," in 2022 IEEE 25th International Conference on Intelligent Transportation Systems (ITSC 2022). IEEE, Oct 2022, pp. 1730–1735.